

ing CPU load. This means that even the Direct3D and OpenGL API layers can be skipped when processing physics, which is a sure-fire way of increasing performance.

Another great advantage ATI has is that their Crossfire cards need not be identical (as SLI forces you to do), and so if you've just bought an X1900XT, and a few years down the line decide to upgrade your graphics card, you need not throw away or sell the X1900XT. All you need to do is force the physics processing on this card, so you save a lot of money and get a very high-end graphics from the newer card at the same time. In fact, ATI claim you need not even be running the cards in Crossfire mode: so long as an ATI card is installed in a PCIe slot and powered, it can take over physics calculations! A boon indeed!

The End Before The Beginning?

No, don't write off Ageia just yet. In order to make completely interactive and realistic worlds, the physics calculation power required will increase *exponentially*! As games get heavier, and system requirement stats soar to all-time highs, most gamers would prefer to have a separate device that does physics computation and also allow them to eke out the maximum from their GPU cards. After all, getting great game physics at just 20 fps is not something we want. We'd rather spend Rs 10,000 or so extra and get great physics at 60 fps!

The bottomline is that Ageia admits that currently, even a GPU will show vast improvements in current physics rendering, but says that once game developers start using advanced collision and fluid dynamic physics in their offerings, the GPU will struggle. Of course there's no telling whether ATI or NVIDIA will revamp their architectures to increase the memory bandwidth, or not! For now, Ageia's PhysX is the only offering that promises integration with upcoming games, and also offers a more realistic feel.

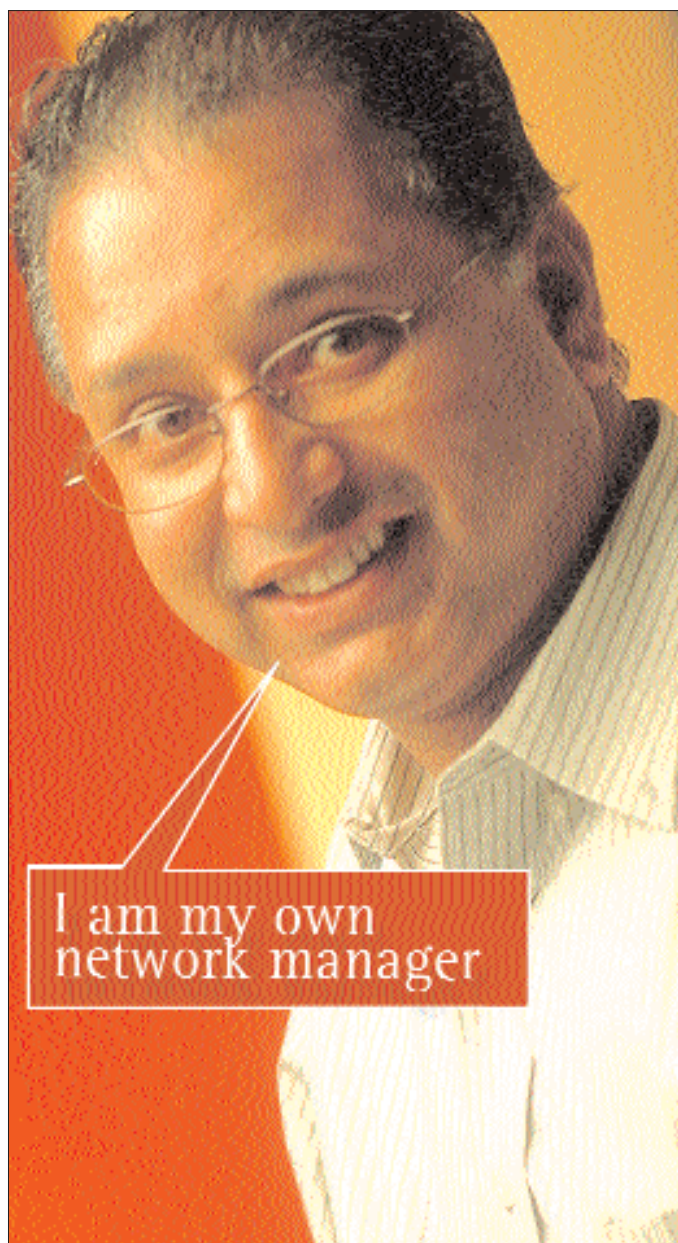
Another point to ponder is that there's no common driver API for PhysX, NVIDIA or ATI, and this is already cause for concern. Game developers are *not* going to make four different versions of a game, one for regular PCs, one for Ageia's PhysX, another for NVIDIA and yet another for ATI. So a standard is needed, and, as usual, everyone's hoping for some intervention from Microsoft here—since most games are made for the Windows platform anyway!

Also, Ageia has a huge head start over NVIDIA and ATI: games have already been developed for its PhysX cards, which are already being retailed, whereas both the graphics giants are only just contemplating the development of methods by which their GPUs can do physics computation.

Whatever the outcome of the battle that's brewing, one thing's for sure: the game industry is set to smash all previous records, and reality will finally enter the game domain! Life has never been this good for us gamers! ■

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Editor's Note: Turn to the next page for a Visual Explanation of how a PPU works!



I am my own
network manager

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